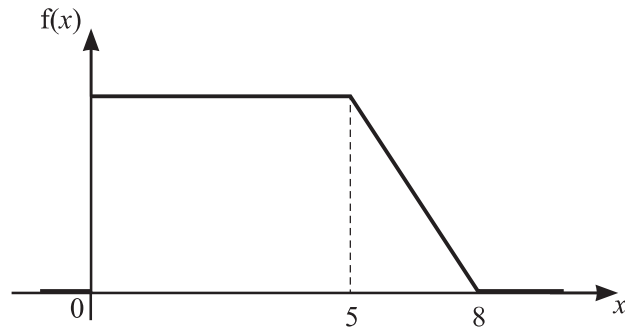


2



As shown in the diagram, the continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} a & 0 \leq x \leq 5, \\ b - cx & 5 \leq x \leq 8, \\ 0 & \text{otherwise,} \end{cases}$$

where a , b and c are constants.

- (a) Show that $a = \frac{2}{13}$ and find the values of b and c . [3]

[illegible]

